Fiscal Stimulus under Sovereign Risk

Bianchi, Ottonello & Presno *JPE*, 131(9), 2023

Presented by Kacper Makowski

The Question

Should a government in recession stimulate or austere when it faces sovereign risk?

- Keynesian answer: always stimulate
- Data: high-risk countries **cut** spending in recessions
- Paper: is procyclicality **optimal**?

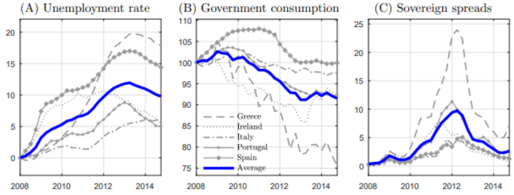

Figure 1: Unemployment, Fiscal Policy, and Sovereign Spreads during the Eurozone Crisis

Core trade-off

Spending ↓ unemployment **but** ↑ default risk and destroys future fiscal space

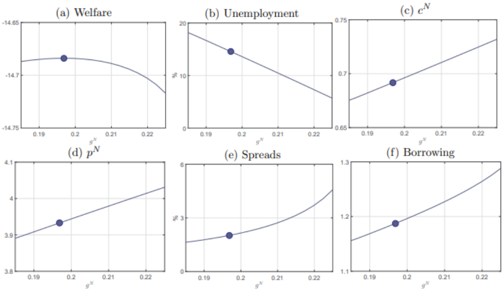

Figure 2: Welfare, Prices, and Allocations with Alternative Spending.

Procyclicality is increasing in sovereign risk

- Higher sovereign risk $\Rightarrow$ more procyclical g
- Higher initial debt $\Rightarrow$ sharper cuts in downturns
- Low-risk countries: closer to Keynesian benchmark

Paper's goal

Rationalize this as **optimal**, not a political failure

The Model

Small open economy, external borrowing by gov, default risk, nominal rigidities, household constraint:

$$P_t^T c_{jt}^T + P_t^N c_{jt}^N = P_t^T y_t^T + \Pi_t^N + W_t h_{jt} - T_t(h_{jt})$$

Nominal rigidities

$W_t \geq \bar{W}$ + fixed e

$\Rightarrow$ unemployment

$\Rightarrow$ multiplier $= 1$

Stimulus **works**

Hand-to-mouth HH

No financial markets

$\Rightarrow$ Ricardian equiv. fails

Imperfect insurance

$\Rightarrow$ unemployment $=$ inequality

Endogenous default

Gov. chooses default each period

$\Rightarrow$ spreads rise *today*

More debt $\Rightarrow$ less future fiscal space

Research design

Blocks 1–2 make stimulus as attractive as possible. Block 3 can still overturn it.

Optimal Policy: Modified Samuelson Rule

$$\underbrace{v'(g) - u_N(c^T, c^N)\Omega(h)}_{\text{Samuelson}} + \underbrace{\mu F'(h)\frac{\partial P^N}{\partial g}}_{\text{Stimulus}} - \underbrace{\eta\left[P^N + \frac{\partial P^N}{\partial g}(g - F(h)\tau)\right]}_{\text{Austerity}} = 0$$

- **Samuelson:** MRS $=$ MRT — standard public goods rule
- **Stimulus** ($\mu > 0$ when rigidity binds): $g \uparrow \Rightarrow P^N \uparrow \Rightarrow$ employment $\uparrow$
- **Austerity** (η): debt financing $\uparrow$ spreads, $\downarrow$ future fiscal space

No default risk

Austerity $= 0$
Stimulus dominates $\Rightarrow$ countercyclical

With default risk

η large at high debt
Austerity dominates $\Rightarrow$ procyclical

Quantitative Results: Calibrated to Spain

Default risk flips the sign of optimal fiscal policy

Table 2: Business Cycle Statistics: Data and Models

Statistic	Data	Model: Risk-free	Model: Default
Averages (in percent)			
mean(spreads)	1.05	0.00	1.09
mean(debt/y)	22.8	22.4	22.6
mean($p^N g^N/y$)	18.1	18.6	18.2
Correlations with GDP			
corr(GDP,g^N)	0.46	−0.81	0.72
corr(GDP,c)	0.98	1.00	0.98
corr(GDP,spreads)	−0.38	0.00	−0.95
corr(GDP,unemployment)	−0.34	−0.37	−0.97
Volatilities (in percent)			
σ(GDP)	3.5	1.2	4.3
$\sigma(p^N g^N)/\sigma$(GDP)	2.0	1.6	2.0
$\sigma(c)/\sigma$(GDP)	1.1	1.1	1.1
σ(spreads)	1.4	0.0	0.7
σ(unemployment)	4.1	0.6	5.6

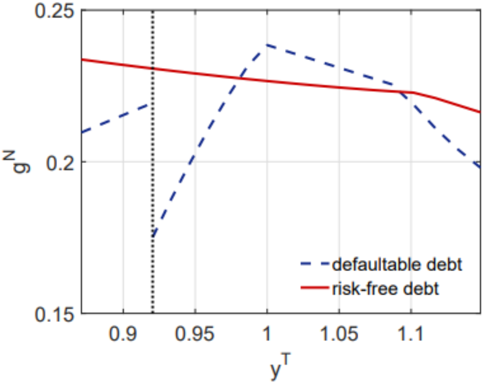

Figure 3: Government Spending as a Function of y^T

State Dependence

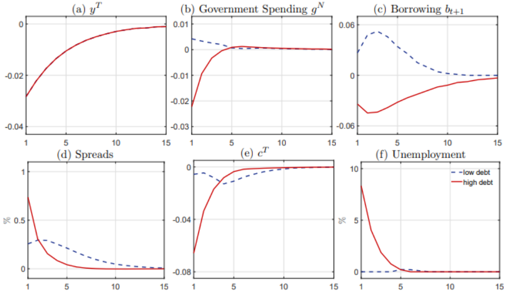

Figure 5: Impulse Responses to y^T for Different Initial Debt Levels

Same recession, different optimal policy:

Low debt

Keynesian gains $>$ sovereign risk cost $\Rightarrow$ **stimulate**

Intermediate debt

Near borrowing limit, spreads spike $\Rightarrow$ **austerity**

Very high debt

Default optimal anyway $\Rightarrow$ **spend post-default**

Fiscal space is the critical state variable

Austerity Can Backfire — Fiscal Forward Guidance

Why austerity backfires:

- $g \downarrow \Rightarrow$ deeper recession $\Rightarrow$ tax revenues $\downarrow$
- Debt/income ratio can *rise* despite cuts
- Default probability may increase

Fiscal Forward Guidance:

- Commit to future surpluses, not current cuts
- $\mathbb{E}_t[\chi_{t+1}] \downarrow \Rightarrow q_t \uparrow$ today
- Stimulus now + credible consolidation later

Key insight

Cut today $\Rightarrow$ demand falls.
Commit to cut tomorrow $\Rightarrow$ spreads fall today.

Empirical Validation

Table 3: Initial Change in Government Consumption and Output during Recessions by Sovereign Risk and External Liabilities

	(1)	(2)	(3)	(4)
Dependent variable	Δg_{jt}	Δg_{jt}	Δy_{jt}	Δy_{jt}
recession	-0.002	-0.003	-0.038^{***}	-0.036^{***}
	(0.005)	(0.006)	(0.005)	(0.005)
recession $\times$ risk	-0.022^{***}	-0.019^{***}	-0.016^{**}	-0.024^{***}
	(0.008)	(0.007)	(0.007)	(0.007)
recession $\times$ risk $\times$ ℓ	-0.008	-0.017^{***}	-0.002	-0.014^{***}
	(0.019)	(0.006)	(0.010)	(0.003)
Observations	1,633	2,368	1,662	2,434
R^2	0.18	0.13	0.45	0.41
Measure of sov. risk	hist. default	min. rating $<$ AA	hist. default	min. rating $<$ AA

- High-risk countries: tighten in recessions, more so at high debt
- Low-risk countries: countercyclical or neutral
- Consistent with model's state dependence

Normative $\rightarrow$ positive

Optimal model matches observed behavior
Procyclicality looks like a **rational response**, not a failure

Summary

What the paper does

1. Documents: procyclicality $\uparrow$ with sovereign risk
2. Model: default + rigidities + HtM households
3. Result: default risk flips optimal policy
4. State dependence: stimulus only at low debt
5. Policy: fiscal forward guidance $\succ$ current austerity

Central message

Procyclical fiscal policy in high-debt countries can be **optimal**.

Not "why ignore Keynes?" but **"what should you do given your constraints?"**

Callibration

Table 1: Calibration

Parameter	Value	Description	Target statistic/Source
Predetermined parameters			
σ	2	Coefficient of risk aversion	Standard business cycle literature
ξ	0.5	Intratemporal elasticity of subst.	Standard business cycle literature
ω	0.3	Share of tradables	Share of tradable GDP (20%)
α	0.75	Labor share in nontradable sector	Uribe and Schmitt-Grohé (2017)
r	0.02	Risk-free rate	Average German 5-year bond return
δ	0.184	Coupon decaying rate	Average bond duration (5 years)
θ	0.18	Reentry probability	Average autarky spell (5.5 years)
κ	0.7	Relative consumption unemployed	Chodorow-Reich & Karabarbounis (2016)
ρ	0.777	AR(1) coefficient of y_t^T	Spanish tradable GDP process
σ_y	0.029	Standard deviation of ε_t	Spanish tradable GDP process
Calibrated parameters			
β	0.91	Discount factor	External debt/GDP (22.8%)
ψ_χ^0	0.33	Utility loss from default (intercept)	Average bond spread (1.05%)
ψ_χ^y	2.42	Utility loss from default (slope)	Volatility of bond spreads (1.4%)
ψ_g	0.02	Weight of g in utility	Relative stdev. of govt. spending to GDP (2)
τ	0.19	Income tax rate	Average govt. spending/GDP (18.1%)
$\bar{w}$	3.1	Minimum wage	Unemployment increase in crisis (10%)